

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Given

- Network Address: **192.168.100.0/24**
- School of Computing: **/25**
- School of Electronics: **/26**
- School of Mechanical Engineering: **/27**

Method Used

The subnets are allocated sequentially from the available /24 network.

Formula used:

Number of IP addresses = $(2^{\{(32-\text{prefix})\}})$

Usable Host Addresses = $(2^{\{(32-\text{prefix})\}}-2)$

Solution

1. School of Computing (/25)

Subnet Mask: **255.255.255.128**

Network Address: **192.168.100.0**

Broadcast Address: **192.168.100.127**

Valid Host Range:

192.168.100.1 – 192.168.100.126

Total IP Addresses:

$(2^7 = 128)$

Usable Host Addresses:

$(128 - 2 = 126)$

Therefore, the School of Computing can support **126 usable hosts**.

2. School of Electronics (/26)

Next available subnet starts at **192.168.100.128**

Subnet Mask: **255.255.255.192**

Network Address: **192.168.100.128**

Broadcast Address: **192.168.100.191**

Valid Host Range:

192.168.100.129 – 192.168.100.190

Total IP Addresses:

$(2^6 = 64)$

Usable Host Addresses:

$(64 - 2 = 62)$

Therefore, the School of Electronics can support **62 usable hosts**.

3. School of Mechanical Engineering (/27)

Next available subnet starts at **192.168.100.192**

Subnet Mask: **255.255.255.224**

Network Address: **192.168.100.192**

Broadcast Address: **192.168.100.223**

Valid Host Range:

192.168.100.193 – 192.168.100.222

Total IP Addresses:

$(2^5 = 32)$

Usable Host Addresses:

$(32 - 2 = 30)$

Therefore, the School of Mechanical Engineering can support **30 usable hosts**.

. Remaining Unused IP Address Range

After allocating the above three subnets, the remaining addresses are:

192.168.100.224 – 192.168.100.255

These addresses are available for future network expansion.

Final Answer

School of Computing (/25)

- Network Address: **192.168.100.0**
- Broadcast Address: **192.168.100.127**
- Valid Host Range: **192.168.100.1 – 192.168.100.126**
- Usable Host Addresses: **126**

School of Electronics (/26)

- Network Address: **192.168.100.128**
- Broadcast Address: **192.168.100.191**
- Valid Host Range: **192.168.100.129 – 192.168.100.190**
- Usable Host Addresses: **62**

School of Mechanical Engineering (/27)

- Network Address: **192.168.100.192**
- Broadcast Address: **192.168.100.223**
- Valid Host Range: **192.168.100.193 – 192.168.100.222**
- Usable Host Addresses: **30**

Remaining Unused IP Address Range

192.168.100.224 – 192.168.100.255

Interpretation

The network has been successfully divided into three subnets according to the given subnet masks. Each department receives an independent subnet with sufficient host addresses, minimizing broadcast traffic and improving network efficiency. The remaining IP address range can be reserved for future departments or additional devices.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Scenario Understanding

A multinational software company has been assigned the private IP block **10.10.0.0/16** for its headquarters. Different departments require different subnet sizes based on the number of employees. The administrator must allocate subnet addresses, determine network and broadcast addresses, identify valid host ranges, calculate usable hosts, and identify the remaining unused IP address range.

Given

Allocated Network: 10.10.0.0/16

Department	Prefix
Software Development	/24
Quality Assurance	/25
Human Resources	/26
Executive Management	/27

Identification of Key Issues

The following tasks are to be completed:

- Allocate subnet addresses sequentially.
- Determine the network address.
- Determine the broadcast address.
- Identify the valid host IP range.

- Calculate the usable hosts.
- Find the remaining unused IP address range.

Application of Concepts

Formula Used

Host Bits

$$= 32 - \text{Prefix Length}$$

Usable Hosts

$$2^{\text{HostBits}} - 2$$

Subnets are allocated from the beginning of the given network.

Department 1 – Software Development (/24)

Subnet Mask

255.255.255.0

Network Address

10.10.0.0

Broadcast Address

10.10.0.255

Valid Host Range

10.10.0.1 – 10.10.0.254

Usable Hosts

$$2^8 - 2 = 256 - 2 = 254$$

Usable Hosts = 254

Department 2 – Quality Assurance (/25)

Next available subnet begins from

10.10.1.0

Subnet Mask

255.255.255.128

Network Address

10.10.1.0

Broadcast Address

10.10.1.127

Valid Host Range

10.10.1.1 – 10.10.1.126

Usable Hosts

$$2^7 - 2 = 128 - 2 = 126$$

Usable Hosts = 126

Department 3 – Human Resources (/26)

Next available subnet begins from

10.10.1.128

Subnet Mask

255.255.255.192

Network Address

10.10.1.128

Broadcast Address

10.10.1.191

Valid Host Range

10.10.1.129 – 10.10.1.190

Usable Hosts

$$2^6 - 2 = 64 - 2 = 62$$

Usable Hosts = 62

Department 4 – Executive Management (/27)

Next available subnet begins from

10.10.1.192

Subnet Mask

255.255.255.224

Network Address

10.10.1.192

Broadcast Address

10.10.1.223

Valid Host Range

10.10.1.193 – 10.10.1.222

Usable Hosts

$$2^5 - 2 = 32 - 2 = 30$$

Usable Hosts = 30

Remaining Unused IP Address Range

After allocating the Executive Management subnet, the remaining address space begins from:

10.10.1.224

Remaining Unused Range

10.10.1.224 – 10.10.255.255

This remaining address space can be reserved for future departments or network expansion.

Decision Making

The subnet allocation follows a logical sequence from the largest department to the smallest. This approach minimizes IP address wastage and provides efficient

utilization of the available **10.10.0.0/16** network while leaving a significant portion of the address space available for future growth.

Final Result

Department	Prefix	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Interpretation

The allocated 10.10.0.0/16 network has been successfully subnetted according to the departmental requirements. Software Development receives 254 usable host addresses, Quality Assurance receives 126, Human Resources receives 62, and Executive Management receives 30. The remaining address space from 10.10.1.224 to 10.10.255.255 is kept available for future expansion, ensuring scalability, efficient IP address utilization, and simplified network administration.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

Scenario Understanding

A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network **2001:0db8:abcd:0000:0000:0000:0000/64**. The task is to compress and expand the IPv6 address, identify the network prefix, and calculate the total number of host addresses available within the subnet.

Given

IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000/64

Identification of Key Issues

The following tasks are to be performed:

- Convert the IPv6 address into compressed notation.
- Expand the compressed address back into full notation.
- Identify the network prefix using the /64 prefix length.
- Calculate the number of host addresses supported within the subnet.
- Determine the total number of possible host addresses available.

Application of Concepts

IPv6 Compression Rules

1. Remove leading zeros from each hexadecimal block.
2. Replace one continuous sequence of all-zero blocks with "::".
3. "::" can be used only once in an IPv6 address.

Step 1: Compress the IPv6 Address

Original Address

2001:0db8:abcd:0000:0000:0000:0000:0000

After removing leading zeros

2001:db8:abcd:0:0:0:0:0

Replace the continuous zero blocks with "::"

Compressed Address

2001:db8:abcd::/64

Step 2: Expand the Compressed Address

Compressed Address

2001:db8:abcd::

Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000

Step 3: Identify the Network Prefix

Prefix Length

/64

The first **64 bits** represent the network portion.

Network Prefix

2001:0db8:abcd:0000::/64

The remaining **64 bits** represent the interface (host) portion.

Step 4: Calculate Host Addresses Supported

Host Bits

$$128 - 64 = 64$$

Total Possible Host Addresses

$$\begin{aligned} &2^{64} \\ &= 18,446,744,073,709,551,616 \end{aligned}$$

Total Host Addresses

18,446,744,073,709,551,616

Approximately

18.4 Quintillion Addresses

Decision Making

Using a **/64 prefix** is the standard practice in IPv6 networks because it provides an enormous address space for end devices while maintaining compatibility with Stateless Address Auto Configuration (SLAAC), Neighbor Discovery Protocol (NDP), and other IPv6 features. This subnet size is ideal for large-scale deployments such as smart hospitals with thousands of connected medical devices and IoT sensors.

Final Result

Task	Result
Original IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000/64
Compressed Address	2001:db8:abcd::/64
Expanded Address	2001:0db8:abcd:0000:0000:0000:0000
Network Prefix	2001:0db8:abcd:0000::/64
Host Portion	Last 64 bits
Total Host Addresses	18,446,744,073,709,551,616 (2⁶⁴)

Interpretation

The hospital's IPv6 network has been correctly compressed to **2001:db8:abcd::/64** and expanded back to its original representation. The **/64** prefix reserves the first **64 bits** for the network and the remaining **64 bits** for host addressing. This configuration supports **18,446,744,073,709,551,616** unique interface addresses, making it highly suitable for future expansion and the deployment of a large number of connected medical devices, IoT sensors, and healthcare systems. IPv6 eliminates address exhaustion while improving scalability, routing efficiency, and network performance.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

Scenario Understanding

An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

- **192.168.1.0/24**
- **192.168.2.0/24**
- **192.168.3.0/24**

A packet arrives at the router with the destination IP address **192.168.2.45**. The objective is to identify the correct destination subnet, matching routing table entry, next-hop interface, and determine the default route if no matching route exists.

Given

Routing Table

Destination Network Prefix

192.168.1.0 /24

192.168.2.0 /24

192.168.3.0 /24

Destination IP Address

192.168.2.45

Identification of Key Issues

The following tasks are to be performed:

- Determine the destination subnet.
- Identify the matching routing table entry.
- Determine the next-hop network/interface.
- Identify the default route if no matching entry exists.

Application of Concepts

Step 1: Determine the Destination Subnet

Given IP Address

192.168.2.45

Subnet Mask

/24 (255.255.255.0)

Applying the subnet mask:

IP Address	:	192.168.2.45
Subnet Mask	:	255.255.255.0

Network Address	:	192.168.2.0

Destination Subnet

192.168.2.0/24

Step 2: Find the Matching Routing Table Entry

Compare the destination network with the routing table.

Routing Entry Match

192.168.1.0/24 No

192.168.2.0/24 **Yes**

192.168.3.0/24 No

Matching Routing Table Entry

192.168.2.0/24

Step 3: Determine the Next-Hop Network/Interface

Since the destination belongs to **192.168.2.0/24**, the router forwards the packet through the interface connected to the **192.168.2.0/24 network**.

Next-Hop Interface

Interface connected to 192.168.2.0/24

(The scenario does not specify an interface name such as GigabitEthernet0/1, so only the connected network can be identified.)

Step 4: Default Route

If the destination IP address does **not** match any routing table entry, the router forwards the packet through the **default route**.

Default Route

0.0.0.0/0

This route is commonly known as the **Gateway of Last Resort**.

Final Result

Task	Result
Destination IP Address	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Routing Table Entry	192.168.2.0/24
Next-Hop Network/Interface	Interface connected to 192.168.2.0/24
Default Route	0.0.0.0/0 (Gateway of Last Resort)

Interpretation

The router successfully identifies that the destination IP address **192.168.2.45** belongs to the **192.168.2.0/24** network. Therefore, it forwards the packet through the interface connected to this subnet. If the destination network were absent from the routing table, the router would use the **default route (0.0.0.0/0)** to forward the packet to the next available router. This routing process ensures efficient packet delivery and reliable communication across interconnected networks.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Scenario Understanding

A company has divided its office network into two separate LANs: **Administration** and **Finance**. Each LAN is connected through a **Layer-3 Router**, which enables communication between different subnets. Based on the given subnet information and router gateway addresses, determine the network details, assign valid IP addresses to hosts, and identify the default gateway for each LAN.

Given Data

Department	Network Address	Prefix	Gateway
Administration LAN	192.168.10.0	/26	192.168.10.1
Finance LAN	192.168.10.64	/26	192.168.10.65

Identification of Requirements

The network administrator must:

- Calculate the network address.
- Determine the broadcast address.
- Find the valid host IP range.
- Assign IP addresses to three computers in each LAN.
- Specify the default gateway for each subnet.

Application of Networking Concepts

Subnet Mask

For /26

255.255.255.192

Block Size

$$256 - 192 = 64$$

Each subnet contains

- Total Addresses = **64**
- Network Address = **1**
- Broadcast Address = **1**
- Usable Hosts =

$$64 - 2 = 62$$

Administration LAN Analysis

Network Address

192.168.10.0

Broadcast Address

192.168.10.63

Valid Host Address Range

192.168.10.1 – 192.168.10.62

Sample Host Assignment

Device	Assigned IP
Computer A	192.168.10.10
Computer B	192.168.10.20
Computer C	192.168.10.30

Default Gateway

192.168.10.1

Finance LAN Analysis

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Address Range

192.168.10.65 – 192.168.10.126

Sample Host Assignment

Device	Assigned IP
Computer A	192.168.10.70
Computer B	192.168.10.80
Computer C	192.168.10.90

Default Gateway

192.168.10.65

Network Summary

Parameter	Administration LAN	Finance LAN
Network Address	192.168.10.0	192.168.10.64
Subnet Mask	255.255.255.192	255.255.255.192
Broadcast Address	192.168.10.63	192.168.10.127
Host Range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Usable Hosts	62	62
Default Gateway	192.168.10.1	192.168.10.65

Decision Making

The company has implemented **subnetting** by dividing the original network into two independent **/26** subnets. A Layer-3 router is used to interconnect both LANs, enabling communication while maintaining separate broadcast domains. Each computer is assigned a unique IP address within its subnet and configured with the appropriate default gateway to ensure successful communication with devices on other networks.

Interpretation

The Administration LAN occupies the subnet **192.168.10.0/26**, while the Finance LAN occupies **192.168.10.64/26**. Each subnet provides **62 usable host addresses**, making them suitable for departmental communication. The configured gateways (**192.168.10.1** and **192.168.10.65**) allow hosts to communicate across different networks through the Layer-3 router. This subnet design enhances network organization, improves performance by reducing broadcast traffic, and provides better scalability and security for future expansion.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand